

PROJECT ARCHITECTURE BLUEPRINT

Rankuno SEO Engine - Phase 1 Page Classification Blueprint

Multi-Dimensional Classification Framework: 3 Interfaces, 6-Signal Consensus Pipeline, ARIA Navigation Trees, CMS Endpoints, and Pydantic Schemas

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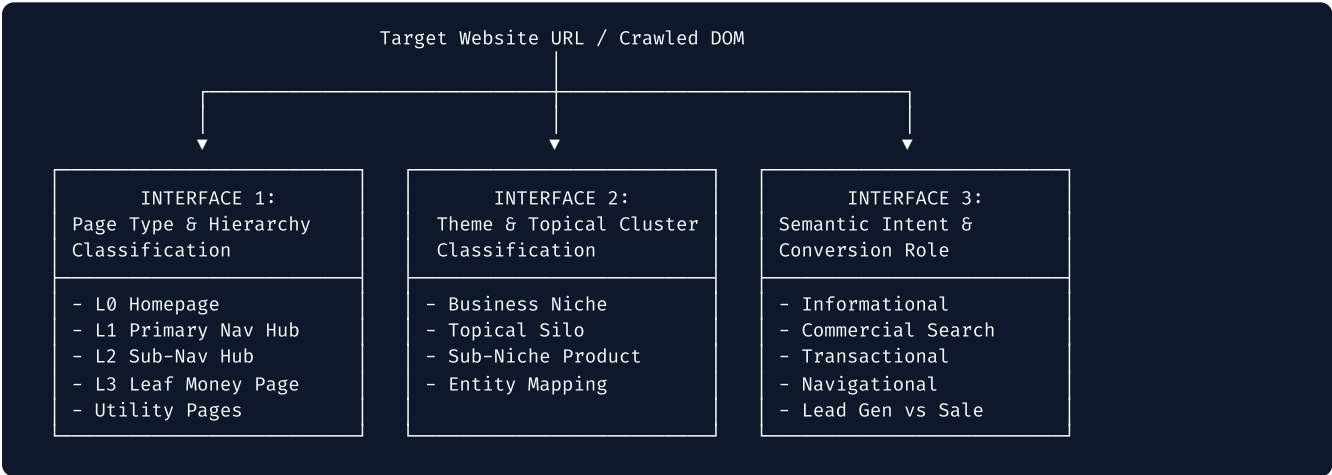
Prepared By: AI Lead

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1. Executive Overview & Purpose

Phase 1 of Rankuno's SEO Engine establishes the **Multi-Dimensional Page & URL Classification Infrastructure**. It categorizes website content graphs across 3 complementary interfaces:



2. Interface 1: Page Type Hierarchy Taxonomy

Hierarchy Level	Level Name	Primary Page Types Included	Example URL Pattern
L0_HOMEPAGE	Root Entry	Homepage	https://vitaquest.com/
L1_PRIMARY_NAV_HUB	Global Parent Hub	Service Category Hub, Blog Hub, Company Hub, Lead Gen Hub	/manufacturing-services/ , /capabilities/
L2_SUB_NAV_HUB	Sub-Category Hub	Sub-Service Category, Product Category Collection, Tag Archive	/manufacturing-services/capsule-supplements/
L3_LEAF_PAGE	Money / Execution Page	Service Detail Page (SDP), Product SKU (PDP), Blog Article	/blog/supplement-manufacturing-trends
UTILITY_PAGE	Infrastructure	Privacy Policy, Terms, Search Results, Faceted Filters, 404s	/privacy-policy , /shop?color=red

3. The 6-Signal Consensus Architecture

To guarantee zero classification failures on complex or flat websites, Rankuno evaluates 6 independent signal streams:

1. **Signal 1: ARIA Navigation Tree Analysis** — Inspects DOM accessibility attributes (`role="navigation"` , `aria-label="Main menu"`) and nested `<ul>/<li>` HTML parent-child structures. Accurately classifies Level 1 vs Level 2 navigation dropdowns even if hidden behind a **Hamburger toggle** (`display: none`).
2. **Signal 2: CMS API Endpoint Footprints** — Queries public CMS data endpoints (WordPress `/wp-json/wp/v2/pages` , Shopify `/collections.json` , `/products.json`). Provides 100% accurate classification for **Flat URLs** (e.g. `site.com/capsules` off root).
3. **Signal 3: Grouped XML Sitemap Index Parsing** — Parses `sitemap_index.xml` to inspect auto-grouped files (e.g. `product-sitemap.xml` , `category-sitemap.xml` , `post-sitemap.xml`).
4. **Signal 4: Schema.org `@graph` JSON-LD Parsing** — Extracts structured schema types embedded in HTML: `CollectionPage` , `ItemPage` , `Service` , `Product` , `Article` , `AboutPage` .
5. **Signal 5: Internal Link In-Degree Centrality** — Evaluates internal link distribution. Pages linked in site-wide headers/footers ($\geq 1,000$ links) are classified as Level 1 Hubs.
6. **Signal 6: LLM Zero-Shot Backup Classifier** — Passes URL, Title, H1, Breadcrumbs, and text snippets to Gemini 3.6 Flash for intelligent fallback classification when zero structural signals exist.

4. Pydantic Data Contracts & Schemas

```

from enum import Enum
from typing import List, Optional
from pydantic import BaseModel, Field

class HierarchyLevel(str, Enum):
    L0_HOMEPAGE = "L0_HOMEPAGE"
    L1_PRIMARY_NAV_HUB = "L1_PRIMARY_NAV_HUB"
    L2_SUB_NAV_HUB = "L2_SUB_NAV_HUB"
    L3_LEAF_PAGE = "L3_LEAF_PAGE"
    UTILITY_PAGE = "UTILITY_PAGE"

class PrimaryPageType(str, Enum):
    HOMEPAGE = "HOMEPAGE"
    SERVICE_CATEGORY_HUB = "SERVICE_CATEGORY_HUB"
    SERVICE_DETAIL_PAGE = "SERVICE_DETAIL_PAGE"
    PRODUCT_CATEGORY_HUB = "PRODUCT_CATEGORY_HUB"
    PRODUCT_DETAIL_PAGE = "PRODUCT_DETAIL_PAGE"
    BLOG_HUB = "BLOG_HUB"
    BLOG_ARTICLE = "BLOG_ARTICLE"
    COMPANY_ABOUT = "COMPANY_ABOUT"
    COMMERCIAL_LEAD_GEN = "COMMERCIAL_LEAD_GEN"
    FACETED_FILTER = "FACETED_FILTER"
    UTILITY_LEGAL = "UTILITY_LEGAL"
    UNKNOWN = "UNKNOWN"

class SearchIntent(str, Enum):
    INFORMATIONAL = "INFORMATIONAL"
    COMMERCIAL_INVESTIGATION = "COMMERCIAL_INVESTIGATION"
    TRANSACTIONAL = "TRANSACTIONAL"
    NAVIGATIONAL = "NAVIGATIONAL"

class SignalScore(BaseModel):
    source: str
    suggested_level: HierarchyLevel
    suggested_page_type: PrimaryPageType
    confidence: float = Field(ge=0.0, le=1.0)
    notes: str

class FullPageIntelligenceProfile(BaseModel):
    url: str
    normalized_path: str
    hierarchy_level: HierarchyLevel
    primary_page_type: PrimaryPageType
    nav_parent_url: Optional[str]
    breadcrumb_path: List[str]
    topical_category: str
    sub_topic: Optional[str]
    search_intent: SearchIntent
    conversion_role: str
    signals_evaluated: List[SignalScore]
    final_confidence_score: float = Field(ge=0.0, le=1.0)
    consensus_method: str

```

5. Edge-Case Handling & Security Audit Matrix

Edge Case / Technical Vulnerability	Root Cause / Impact	Rankuno Architectural Guardrail
Hamburger / Hidden Mobile Navs	CSS hides navigation (<code>display: none</code>).	Parse DOM ARIA tree (<code>role="navigation"</code>) ignoring visual CSS display.
Flat URLs (<code>site.com/capsules</code>)	Directory path depth regex fails.	Query CMS Endpoints (<code>/collections.json</code> , <code>/wp-json/</code>) & Sitemap index files.
JavaScript SPAs (React/Next.js)	Empty <code><div id="root"></code> returned.	Playwright headless browser rendering fallback to hydrate dynamic DOM.
Multi-Language Routing (<code>/en/</code> , <code>/es/</code>)	Locale prefixes skew category matching.	Path Normalizer: Strip locale prefixes (<code>/en/services</code> → <code>/services</code>) prior to matching.
Faceted Filter Query Params	Filter parameters (<code>?color=red</code>) create infinite URLs.	Parameter Normalizer: Group query parameter URLs into <code>FACETED_FILTER</code> utility type.
Anti-Bot & Cloudflare Rate Limits	IP blocks during bulk scraping.	Rotate User-Agent headers, enforce backoff retries, and prefer CMS API endpoints.
API Cost Overflow	Unbounded LLM fallback API calls.	Rate-limiter & caching layer: LLM invoked only when 5 structural signals are ambiguous.

6. Strategic Implementation Roadmap

Phase 1: Multi-Dimensional Page Classification Engine

[CURRENT PHASE]

1A: Blueprint, Schemas & Edge-Case Audit (Sign-off)

1B: Core Signal Parsers (ARIA, CMS Client, Sitemap, Schema)

1C: Consensus Engine & LLM Zero-Shot Backup

1D: Evaluation Harness, Automated Verification & Unit Test Suite

Phase 2: Technical SEO Crawling & Audit Engine

Dynamic JS Crawler & DOM Parser (Playwright)

Core Web Vitals (PageSpeed Insights API Integration)

Automated Technical Audit Reporter

Phase 3: Keyword Intelligence & Vector Clustering Engine

Semantic Embeddings & Intent Classifier

Content Gap & Cannibalization Analyzer

Phase 4: Paid Search (PPC) Campaign & Ad Engine

Automated RSA Copy Generator & Negative Keyword Finder

PPC Performance Anomaly Sentinel

Phase 5: Market Research & Competitor Intelligence Engine

SERP & Competitor Content Gap Scraper

Automated Executive & Client Performance Reporter

Phase 6: Autonomous Multi-Agent Orchestration & Command Dashboard

Multi-Agent Execution (SEO Agent + PPC Agent + Research Agent)

Executive Analytics & Web Command UI

7. Implementation Gap Register & Risk Register

#	Gap	Standard §	Priority
1	No Dockerfile / docker-compose.yml	§DevOps	High (Phase 2)
2	No evaluation harness or golden set for classifier	§Evals	High (Phase 1D)
3	Coverage floor configured (85%) but not blocking CI	§Quality	High